LaserCortex

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Chapter 1

EML Registry: Tamari Lattice Contraction

1.1 Core Types

Definition 1 (EML Tree)**.** The type of binary trees representing configurations in the Tamari lattice. A tree is either a `Leaf` (empty configuration) or a `Node` t_1 t_2 (composition of two sub-configurations).

Definition 2 (Tree Size)**.** The number of internal nodes in a tree, used as the lattice index n in T_n.

Definition 3 (Single-Step Contraction)**.** The primitive non-associative rewrite rule: $(a \bullet b) \bullet c \to a \bullet (b \bullet c)$. This is the coupling signature for choice resolution, lifted through subtrees.

Definition 4 (Multi-Step Contraction)**.** The reflexive-transitive closure of $\mathsf{contracts\_one}$. Represents an audit trail of choice resolutions with monotonic provenance.

Definition 5 (Right-Comb Normal Form)**.** The minimum element of the Tamari lattice — the equilibrium attractor. Defined as $\mathsf{rightComb}(0) = \mathsf{Leaf}$ and $\mathsf{rightComb}(n{+}1) = \mathsf{Node}(\mathsf{Leaf},\ \mathsf{rightComb}(n))$.

1.2 Lifting Lemmas

Lemma 6 (Contraction Preserved Under Left Node)**.** *If $t \to t'$ then $\mathsf{Node}(t,\ r) \to \mathsf{Node}(t',\ r)$. Choice resolution in the left subtree propagates to the whole composition.*

Lemma 7 (Contraction Preserved Under Right Node)**.** *If $r \to r'$ then $\mathsf{Node}(l,\ r) \to \mathsf{Node}(l,\ r')$. Choice resolution in the right subtree propagates to the whole composition.*

Lemma 8 (Transitivity of Contraction)**.** *If $s \to t$ and $t \to u$ then $s \to u$. Audit trails concatenate monotonically.*

1.3 Main Contraction Theorem

Lemma 9 (Node of Right-Combs Contracts to Right-Comb)**.** *For any $a, b : \mathbb{N}$,*

$$\mathsf{Node}(\mathsf{rightComb}(a),\ \mathsf{rightComb}(b)) \longrightarrow \mathsf{rightComb}(1 + a + b).$$

The equilibrium attractor is closed under non-associative composition.

Theorem 10 (Every Tree Contracts to Right-Comb)**.** *For any tree $t : \mathsf{EMLTree}$,*

$$t \longrightarrow \mathsf{rightComb}(t.\mathsf{size}).$$

Every configuration has a valid evolution path to the equilibrium attractor indexed by its size.

1.4 Router and Registry

Definition 11 (Router Index)**.** A bounded natural number $\mathsf{Fin}(n)$ serving as a neural binding address.

Definition 12 (Type Registry)**.** An injective mapping from router indices to EML trees. This is the cortex-registry interface where neural computation meets formal grammar.

1.5 Witness-Skeptic Game

Definition 13 (Cortex Certificate)**.** A proof-carrying audit trail $\langle source, target, proof \rangle$ showing that a tree reaches its equilibrium.

Definition 14 (Certify)**.** Issues a $\mathsf{CortexCertificate}$ for any tree t, certifying that $t \to \mathsf{rightComb}(t.\mathsf{size})$. This is the quench witness in the Witness-Skeptic game.

Chapter 2

Logic Types: Pluralistic Reasoning Framework

2.1 Logic Type Hierarchy

Definition 15 (13 Logic Types)**.** An inductive type enumerating 13 reasoning paradigms: Fuzzy, ManyValued, Paraconsistent, Temporal, Deontic, Epistemic, Quantum, Intuitionistic, Relevance, Free, Infinitary, Modal, and Classical.

Each handles specific classes of paradoxes as boundary conditions rather than errors.

Definition 16 (Logic Classification)**.** Classification of logic types as Substructural, Extension, Alternative, Generalization, or Classical.

2.2 Logic-Specific Contraction

Definition 17 (Logic Tree)**.** A type family assigning EML trees to each logic type. Currently uniform; extensible to logic-specific tree types.

Definition 18 (Logic Contraction)**.** A type family of contraction relations indexed by `LogicType`. For Classical logic this is the Tamari contraction from EMLRegistry.

2.3 Meta-Contraction

Definition 19 (Meta-Contraction)**.** Cross-logic contraction allowing reasoning that spans multiple logic types via intra-logic contraction, inter-logic translation, and transitivity.

Definition 20 (Logic Translation)**.** A structure specifying forward and backward maps between logic types with soundness and completeness proofs.

2.4 Cayley-Dickson Connection

Definition 21 (CD Step)**.** Maps logic types to Cayley-Dickson construction steps: Classical $\to 0$, Fuzzy $\to 1$, Intuitionistic $\to 2$, Quantum $\to 3$, Paraconsistent $\to 4$.

Definition 22 (Split-Octonion Sectors)**.** Divides logic types into associative (e_0–e_3) and non-associative (e_4–e_7) sectors of the split-octonion algebra with $(4,4)$ signature.

2.5 Normal Forms and Contraction Theorem

Definition 23 (Logic Normal Form)**.** The normal form for each logic type. For Classical logic this is rightComb.

Theorem 24 (Logic Contraction to Normal Form)**.** *For any logic type lt and tree t,* LogicContraction lt t (LogicNormalF
Currently proven for Classical logic; other cases are placeholders.